

Test 2

November 20, 2023

Problem 1

Solve the inequality, graph the solution, and give interval notation;

$$\frac{m}{5} \leq -\frac{6}{5}.$$

Problem 2

Solve the compound inequality, graph the solution, and give interval notation;

$$\frac{v}{8} > -1 \text{ and } v - 2 < 1.$$

Problem 3

Solve the inequality, graph the solution, and give interval notation;

$$3 - 2|x - 5| \leq -15$$

Problem 4

Solve the system of equations;

$$-2x + 2y = 18$$

$$y = 7x + 15.$$

Problem 5

Solve the system of equations;

$$-x - 5y = 28$$

$$-x + 4y = -17.$$

Problem 6

Solve the system of equations;

$$x + y + z = 6$$

$$2x - y - z = -3$$

$$x - 2y + 3z = 6.$$

Problem 7

A collection of dimes and quarters is worth \$15.25. There are 103 coins in all. How many of each is there?

Problem 8

A tank contains 8000 liters of a solution that is 40% acid. How much water should be added to make a solution that is 30% acid?

Problem 9

Simplify:

$$\frac{2x^7y^5}{3x^3y \cdot 4x^2y^3}.$$

Problem 10

Simplify:

$$\frac{y^3 \cdot x^{-3}y^2}{(x^4y^2)^3}.$$

Problem 11

Simplify in scientific notation:

$$\frac{5.33 \times 10^{-6}}{9.62 \times 10^{-2}}.$$

Problem 12

Simplify: $(5n^4 + 6n^3) + (8 - 3n^3 - 5n^4)$.

Problem 13

Simplify: $(6n - 4)(2n^2 - 2n + 5)$.

Problem 14

Simplify: $(4v - 7)(4v + 7)$.

Problem 15

Divide:

$$\frac{3n^3 + 9n^2 - 64n - 68}{n + 6}.$$

Problem 16

Simplify: $(2a^4)^4$.

Problem 17

Divide:

$$\frac{x^3 - 16x^2 + 71x - 56}{x - 8}.$$

Problem 18

There were 200 tickets sold for a women's basketball game. Tickets for students were \$0.50 each and for adults \$0.75 each. The total amount of money collected was \$132.50. How many of each type of ticket was sold?

Problem 19

Solve the inequality, graph the solution, and give interval notation; $6 - 3|1 - 4x| < -3$.

Extra Credit

Solve the system of equations;

$$w + x + y + z = 2$$

$$w + 2x + 2y + 4z = 1$$

$$-w + x - y - z = -6$$

$$-w + 3x + y - z = -2.$$